Analyse and Gather Twitter Data Using Apache Flume

Apache Flume is a service for streaming logs into Hadoop. It is a distributed and reliable service for efficiently collecting, aggregating and moving large amounts of streaming data into the Hadoop Distributed File System (HDFS). In this article, the authors adapt Flume for analysing and gathering data from Twitter.

Hadoop is an open source Apache framework based on Java, and is mainly used to store and process very large data sets of computer clusters. A Hadoop framework application can be scaled up to thousands of machines, each offering its own local computation and storage. There are two parts to the Hadoop framework, which is considered to be the core. One is HDFS, which stands for Hadoop Distributed File System — this is the storage part, The other is MapReduce, which takes care of data processing. Listed below are the modules that are part of the Hadoop framework,

- Hadoop Common
- HDFS – Hadoop Distributed File System
- Hadoop Yarn
- Hadoop MapReduce

Challenges with data load in the Hadoop framework

Once the data gets into the Hadoop framework, you have won half the battle but the real struggle begins when new, fast-moving data streams starts flowing in. For example, system or application logs, geolocation data and monitoring, as well as social media trends and updates — all constitute fast-moving streams requesting for storage in the Hadoop Distributed File System (HDFS). Hence, the need arose for creating a standard on how all the data streams could be directed into the HDFS more efficiently. That's where the evolution began and, currently, there are many solutions in place and the most widely used one being Apache Flume.

What is Apache Flume?

Flume is an extremely reliable service, which is used to efficiently collect, aggregate and store large amounts of data. Multiple Flume agents can be configured to collect high volumes of data. It also provides a simple extensible data model that can be used for any analytic application. Some of the features of Apache Flume are given in Table 1.

Table 1

Feature	Description
Streaming data and aggregator	This feature allows Flume to take in streams from multiple sources and then push them onto HDFS for storage and analysis
Contextual routing and it acts as a buffer	This feature helps it act as a mediator when the incoming data rate increases, and provides a steady flow of data. Also, it provides contextual routing.
Reliable and efficient data delivery	Flume uses channel-based transactions to guarantee reliable message delivery. It uses channel-based delivery, where there is always one sender and one receiver contained for every message.
Scales horizontally	Has the ability to take in new data streams and additional storage volume, on the fly.

This article illustrates how quickly you can set up Flume agents to collect fast-moving data streams and to push the data into Hadoop's file system from Twitter. By the time we're finished, you should be able to configure and launch a Flume agent and understand how various data flows are easily constructed from multiple agents.

Setting up Flume to analyse Twitter data

The first step is to install Flume. Here, we are using CENT OS 6.7 in a virtual machine where we are going to deploy it.

Download Flume using the following command: